Answers and notes

Assignment 2 - TDT4240

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Step 1

I chose the pong game from last time.

Step 2

I converted the Ball class to follow the Singleton pattern. I thought this class would fit this pattern best as the game should only have one ball.

Step 3

I chose to implement the Model View Controller pattern.

Step 4

a)
- Architectural patterns
- Pipe & Filter
- Model View Controller
- Entity Component System
- Design patterns
- Observer pattern
- State pattern
- Abstract
- Template

The main difference between Architectural Patterns and Design Patterns, is that the Architectural Patterns involves the entire software system and defines predefined subsystems with their responsiblities and rules for how the different subsystems should communicate. While Design Patterns addresses how to refine subsystems/components, or the relationships between them, in a specific programming language.

Architerctural Patterns can be seen as more generalised Design Patterns, since they are developed to work on more systems.

b)
Since I chose the MVC pattern, I had to split my program into three different parts. View, Controller, and Model.

I converted the main class `MyGdxGame` into the view, and it's responsibility is to draw everything on screen.

The controller classes are found inside the `controllers` folder. `Controller` is the main class and takes inputs from the user and converts them into movement. I also created the `BallController` and `PaddleController` classes to seperate out some functionality.

Lastly, the models which are found in the `models` folder. These represent the Ball and Paddles in the game, and include very little logic. They are only moved when the controller tells them to.

c)
The main advantage is that the code has become a lot more readable than before, and it's a lot easier for new developers to do changes to the program since they know that each part of the program does.

One of the disadvantages is that it takes a lot more time to create the program, since you have to follow a set of rules. In a program as small as this it is not necessary to use any fancy patterns, but I see the use of it in larger projects.